Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/30/2023
PI Name:	Doherty, Colleen Ph.D.		
Project Title:	From the Perspective of the Target, a Cis-Regulatory Perspective on the Effects of Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	11/01/2018	**End Date:**	10/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2022 per NSSC information (Ed., 10/5/21) NOTE: End date changed to 10/31/2021 per NSSC information (Ed., 9/17/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Perera, Imara Ph.D. (North Carolina State University)		
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Task Description:	Long duration spaceflight and future colonization missions will require plants to provide much needed resources for human health and well-being. Plants are amazingly hardy organisms which have over the years, developed mechanisms to perceive and respond to stresses they encounter. This evolution occurred in terrestrial environments here on Earth, and not in the environment of space. We are interested in understanding the mechanisms by which plants perceive and respond to stresses, such as a change in gravity or other aspects of spaceflight. Understanding how plants sense and respond to this environment is critical for enabling plant growth in space. We examined the available transcriptional data in GeneLab and observed that, while there were few genes that were cohesively expressed across all conditions, there was some consistency in the cis-regulatory features that are enriched. Based on findings from GeneLab Transcriptional experiments we propose to examine the effects of spaceflight through the lens of the cis-regulatory network. Our analysis of the Arabidopsis transcriptional datasets available in GeneLab shows that there is little overlap in the individual differentially expressed (DE) genes between the various spaceflight experiments. Rather than querying the overlap of specific genes across experiments, we are focusing on upstream signaling components, using the conserved cis-regulatory sequences as a tool to uncover the upstream regulators. We have identified enriched and conserved motifs in the promoters of genes differentially expressed in spaceflight experiments. This approach is inclusive and particularly beneficial for identifying patterns in limited and highly dissimilar data sets. We will take a focused approach, selecting a few of the motifs that are enriched in genes that are responsive to microgravity across multiple independent experiments in GeneLab and examine the cis-regulatory control from the bottom up. The objective of this study is to utilize the GeneLab repository as a starting point to identify mechanisms regulating plant responses to spaceflight and microgravity. Our specific aims are: 1. To screen a yeast-one-hybrid (Y1H) transcription factor library to identify specific transcription factors This objective is to screen the three motifs indicated in the proposal and simultaneously identify additional motifs, integrating additional data available from GeneLab, particularly those motifs enriched in the 5’ Untranslated Region (UTR) and first intron. 2. Confirm DNA-binding of transcription factors and obtain transcription factor mutants This objective is to test the identified candidate transcription factors that bind to the potential gravity responsive elements and confirm their binding to the promoter region in yeast and test in planta (objective 3) their effects on the induction of the reporters generated from these elements. 3. Generate promoter:reporter lines to test in planta In this objective we propose to develop reporter constructs for our top candidates in planta to test the response of these motifs. Modifications to these objectives, in part due to the benchwork aspects of this project being significantly disrupted due to COVID-related lab restrictions and closures, included the following aims for the project: 4. Develop improved methodologies for identifying conserved motifs from diverse data sets, test these on synthetic data sets and other stresses, and optimized methods to microgravity data from GeneLab 5. Identify candidate polymorphisms from RNA-Seq data to evaluate the potential effects of Radiation-induced DNA damage on organisms in the ISS from RNA-Seq data available on GeneLab. This proposal is in response to Appendix G of the Research Opportunities in Space Biology research announcement. Specifically, our proposal will address 4b) Systems Biology Experimental Research using the GeneLab Data System under "Investigators who wish to develop new experimental hypotheses based on data in the GeneLab database." We anticipate that this work will enable us to delineate signaling pathway/s responding to microgravity and yield new knowledge on plant specific responses to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research provides a fundamental understanding of how plants respond to environmental stimuli. Our cis-regulatory network and new approach developed in this proposal can be applied to stresses beyond gravity to identify the regulatory elements that enable a plant to sense and respond to a given stress. Our new network analysis approach is focused on the cis-regulatory elements and provides a novel perspective to stress responses. While this analysis method is applied to identifying microgravity-responsive regulatory features, it can be expanded to identify regulatory elements driving responses to other stresses, which can help improve plant growth in extreme conditions on Earth or off-planet. The two new data analysis approaches developed in this project have applications to both Space Biology and benefit life on Earth. The pipeline developed here to identify polymorphisms from RNA-seq data can be applied to the vast amount of RNA-Seq data available in GeneLab to identify the effects of space radiation on nucleotide changes. The improved method of identifying cis-regulatory components can be applied to identify microgravity-responsive transcripts for improved understanding of the effects of space travel on organisms. However, it can also be applied to identify cis-regulatory features enriched in transcriptional responses to change in environments on Earth such as those induced by biotic or abiotic stressors.
	Microgravity Motif Identification We developed a new method for determining cis-regulatory motif candidates that regulate changes in gene expression in response to altered gravity levels. This method uses machine learning classification to identify motifs that are predictive of the response observed. The test case is if the motif is altered in microgravity or not. We completed the pipeline optimization and developed user-friendly tools to facilitate the pipeline application. We added benchmarking against synthetic data sets and additional benchmarks against other co-regulated gene responses to abiotic and biotic stress. These benchmarks provided insights into the best metrics for optimizing the algorithm and tuning parameters essential for addressing that optimization. Tuning to optimize the novel metrics involved optimizing a different set of parameters than the more traditional approaches of optimizing to the Area Under the receiver operating characteristic (ROC) Curve (AUC).

Task Progress:	We have completed the initial in planta evaluations of the candidates identified before improvement with the synthetic data set and new parameter turning. These demonstrate a response to the conditions they were identified in for cold, heat, and biotic responses. For microgravity, we see a weak response in hypergravity and a highly variable response in simulated microgravity. The results are consistent with our predictions, but we hope that the candidates identified with our optimized approach will have stronger signals. Identifying Variants Using RNA-Seq Data This research has been completed and a manuscript has been submitted for peer review and is available on bioRxiv (detailed in the Bibliography). The pipelines for both approaches are available in GitHub (detailed in the Bibliography).
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Knight MS, Nielsen DM, Doherty CJ. "Assessing the nucleotide-level impact of spaceflight stress using RNA-sequencing data." 2022 Annual Meeting, Southern Section of the American Society of Plant Biologists, Birmingham, AL, March 26-28, 2022. Abstracts. 2022 Annual Meeting, Southern Section of the American Society of Plant Biologists, Birmingham, AL, March 26-28, 2022. , Mar-2022
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Abstracts for Journals and Proceedings	Knight MS, Nielsen DM, Doherty CJ. "Linking essential cis regulatory information to gene expression patterns using machine learning. " 2022 Genetics and Genomics Association Event, Raleigh, NC, November, 2022. Abstracts 2022 Genetics and Genomics Association Event, Raleigh NC, November, 2022. , Nov-2022
Articles in Other Journals or Periodicals	Knight MS, Desai JS, Nielsen DM, Doherty CJ. "MotifBoostR" GitHub server. Posted October 17, 2022. https://github.com/montana-knight/motifboostR , Oct-2022
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Dissertations and Theses	Knight MS. "Advanced applications of RNA-sequencing data in plant bioinformatics." Dissertation, North Carolina State University, October 2022. , Oct-2022